0109026/27 2&1 B.B.E.K.O
September 2026
MATHEMATICS 2&1 **2 &1**
Essay & Objective
2 hours

Name:.....

Index Number:.....

BEST BRAIN EXAMINATIONS KONSORTIUM
GHANA

Special Private Mock Examinations for BECE Candidates

September 2026 **MATHEMATICS 2 & 1** 2 hours

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name** *and* **index number** *in* **ink** *in the spaces provided above.*

This booklet consists of **two** *papers. Answer Paper* **2** *which comes first, in your answer booklet and Paper* **1** *on your Objective Test answer sheet. Paper* **2** *will last* **1** *hour after which the answer booklet will be collected. Do* **not** *start Paper* **1** *until you are told to do so. Paper* **1** *will last* **1** hour.

The use of calculators is **not** *allowed.*

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PAPER 2

1 hour

Essay

Answer **four** *questions* **only.**

All questions carry equal marks.

All *workings must be clearly shown. Marks will* **not** *be awarded for* **correct** *answers without corresponding working.*

1. (a) The table shows the sales made by a trader in 3 days.

Day	Sales (GH¢)
Tuesday	700
Wednesday	1,000
Friday	1,250

Write, simplifying the results, the ratio of the sales made on;

(i) Tuesday to Friday;

(ii) Wednesday to Tuesday;

(iii) Friday to total sales in the 3 days.

(b) Solve the following equations:

(i) $30 = 16 - 2x$;

(ii) $37 - 8w = 2w - 13$

(c) Fiifi, Ama and Yaa start to jog around a circular track at the same time. They complete their rounds in 36 seconds, 48 seconds and 42 seconds respectively. After how many seconds will they be together at the starting point?

(d) Express 0.05% as a proportion of 200km.

2. (a) Mr. Okoe's annual salary as an examinations coordinator is GH¢ 42,000.00. If 40% of his annual income goes to his children's school fees and 10% goes for rent, calculate

(i) cost of school fees;

(ii) cost of rent;

(iii) balance remaining after 10% tithe deduction.

(b) Simplify $4x + 2 + 8 > 30$

(c) A typist can type 100 words in 4 minutes. How many words can she type in

(i) 1 minute?

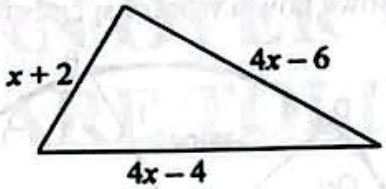
(ii) 5 minutes?

(d) Osei has a rectangular field. It is 85m long and 25m wide. How long is the fence round the field?

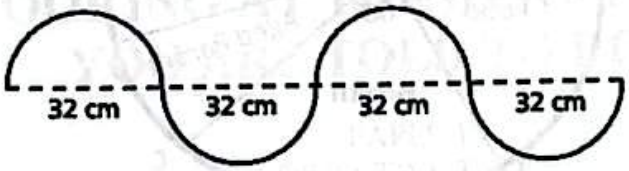
3. (a) Given that $m = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $n = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $r = \begin{pmatrix} 18 \\ -6 \end{pmatrix}$. Find $m + n + r$

(b) If the perimeter of the triangle below is 46 *mm*, find the dimensions of the triangle.

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(c) A wire is bent to form four semi-circles. How long is the wire? [$\pi = 3.14$]



(d) A salesman sells a mean of 4.6 solar power systems per day for 5 days. How many must he sell on the sixth day to increase his mean to 5 sales per day?

4. (a) (i) Complete the table using the rule, $y = -2x + 4$

x	- 3	- 2	- 1	0	1	2	3	4
y			6		2		- 2	

(ii) Using a scale of 2cm to 1 unit on the x – axis and 2cm to 2units on the y- axis, plot 2 perpendicular axes **OX** and **OY** on a graph sheet. Label the x – axis from – 3 to 5 and y – axis from – 6 to 12.

(iii) Use the completed table to plot the points on the graph and draw a straight line through the points, using a straight edge.

(b) Express 48 hours in seconds, leaving your answer in words.

(c) Evaluate; [5(7 – 3) – 4 (5 – 8)]

(d) Round off **48.293847** to the nearest **tenth**, **hundredth** and **thousandth**

5. (a) $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are position vectors relative to the origin O. Given the points $A(3, 1)$ and $B(-1, -2)$

(i) write down $\overrightarrow{OA}$ and $\overrightarrow{OB}$ as column vectors;

(ii) Express $\overrightarrow{AB}$ as a column vector;

(iii) Calculate the magnitude of $\overrightarrow{AB}$.

(b) Simplify the following:

(i) $5\frac{3}{4} - 2\frac{7}{8} + 1\frac{1}{2}$;

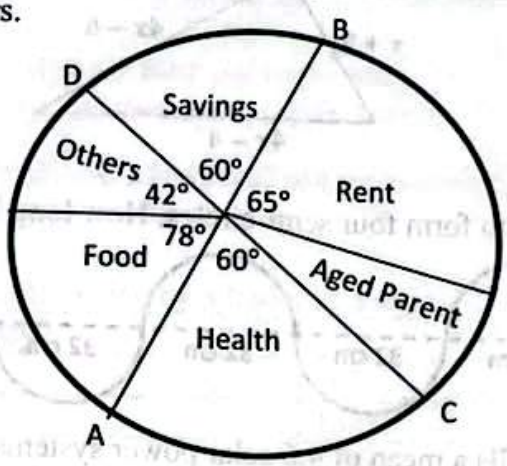
(ii) $\frac{8}{9} \times \frac{4}{6}$

(c) A man gave $\frac{1}{3}$ of his salary to his daughter, $\frac{1}{5}$ to his first son and $\frac{2}{5}$ to his wife. What fraction of his salary is left?

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(d) The pie chart below shows how a worker spends his monthly salary. The lines AB and CD are the diameters.



If he spends GH¢ 66.00 on his aged parent, find,

(i) how much he earns in a month;
(ii) how much he spends on others;
(iii) the percentage of his salary spent on health and food.

6. (a) The students in a class were asked to state the number of children in their families. Their responses are given below.

1, 2, 1, 3, 2, 1, 2, 4, 2, 2, 1, 3, 1, 2,
2, 2, 1, 1, 6, 3, 1, 2, 1, 2, 2, 1, 2, 3

(i) Draw a frequency table to represent this information.
(ii) Calculate the average number of children the students have in their families.

(b) If $r = \frac{3}{4}$ and $s = -\frac{1}{3}$, find q when;

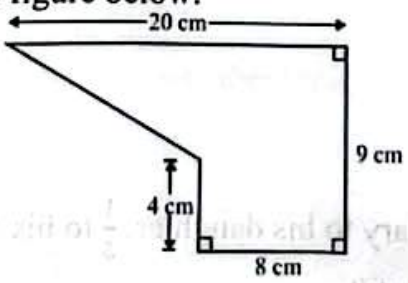
(i) $q = r + s$
(ii) $q = 3s + 2r$

(c) The table below shows the number of customers a new restaurant had each day.

Day	2	3	4	5	6
Customers	7	9	11	13	15

(i) Write a pattern rule to calculate the number of customers on any day.
(ii) Use your pattern rule to determine the number of customers on day 11.
(iii) When will the restaurant receive 23 customers?

(d) Find the area of the figure below.



END OF ESSAY TEST

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Answer **all** *questions*
Each question is followed by **four** *options* **A** *to* **D**. *Find out the* **correct** *option for* **each** *question and shade* **in pencil** *on your answer sheet, the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

What is the smallest number which is divisible by 16 and 20?
A. 80
B. 40
C. 120
D. 160

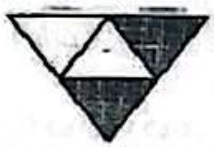
The correct answer is 80, *which is lettered A and therefore answer space* **A** *would be shaded.*

[A] [B] [C] [D]

Think carefully before you shade the answer spaces. Erase completely an answer you wish to change.
Do all rough work on this question paper.
Now answer the following questions

1. The sum of 7 and 5 times of a number is 47. Find the number
A. 6
B. 7
C. 8
D. 9

2. What is the simplest fraction of the shaded portion of the figure below?
A. $\frac{2}{3}$
B. $\frac{1}{2}$
C. $\frac{1}{4}$
D. $\frac{1}{3}$



3. The bearing of Ati from Lomnava is 130°. Find the bearing of Lomnava from Ati.
A. 40°
B. 220°
C. 310°
D. 130°

4. Which of the following statements is **false**?
A. $2 + 1 = -1$
B. $-5 + 1 = -4$
C. $2 + (-1) = 1$
D. $8 + (-9) = -17$

5. Order the fractions $\frac{1}{3}, \frac{2}{5}, \frac{3}{8}, \frac{7}{10}$ from the smallest to the largest.
A. $\frac{1}{3}, \frac{2}{5}, \frac{3}{8}, \frac{7}{10}$
B. $\frac{3}{8}, \frac{2}{5}, \frac{1}{3}, \frac{7}{10}$
C. $\frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{7}{10}$
D. $\frac{2}{5}, \frac{1}{3}, \frac{3}{8}, \frac{7}{10}$

Use the information below to answer questions **6** *to* **8**

$P = \{3, 6, 9, 12, __, __\}$
$Q = \{1, 2, 3, 6, 9, 18\}$

6. Which of the following best describes set **Q**?
A. Multiples of 2
B. Factors of 21
C. Prime numbers up to 18
D. Factors of 18

7. Find the missing elements in set **P**
A. 13, 15
B. 15, 18
C. 14, 16
D. 15, 19

8. Find $P \cap Q$
A. {3. 6. 9}
B. {3, 6, 9, 15}
C. {3, 6, 9, 18}
D. {3, 6, 9, 15, 18}

9. Find the image of the point (- 2, 7) under a reflection in the y – axis.
A. (- 2, 7)
B. (2, - 7)
C. (- 2, - 7)
D. (2, 7)

10. Find the product of 0.025 and 1.5
A. 3.75×10^{-2}
B. 3.75×10^{2}
C. 0.375
D. 3.75

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11. Simplify $1\frac{1}{5} \div 2\frac{1}{10}$

A. $2\frac{13}{25}$

B. $\frac{12}{63}$

C. $\frac{25}{63}$

D. $\frac{4}{7}$

12. The angles of a triangle are in the ratio 5:3:2. Find the value of the highest angle.
A. 90°
B. 120°
C. 180°
D. 360°

13. By how much is 7,249,004 greater than 7,248,261?
A. 473
B. 347
C. 437
D. 743

14. Find the product of 5.65×10^4
A. 56,500
B. 5.60000
C. 56.500
D. 5,60000

15. Simplify $\frac{2}{3}(x-y)+x-2y$

A. $\frac{5}{3}x-\frac{8}{3}y$

B. $\frac{4}{3}x-\frac{7}{3}y$

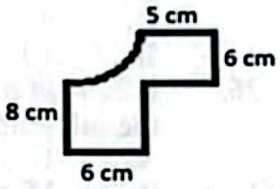
C. $\frac{5}{2}x-\frac{8}{3}y$

D. $\frac{5}{3}x-\frac{6}{3}y$

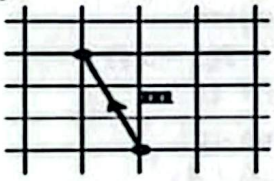
16. The question "*What do you think about the new hall*" is
A. an open question.
B. a closed question.
C. a tall question.
D. a short question.

17. If $p = 3x - 4y - 8z$, $q = -10 + 7x + 11z$ and $r = 19z - 6y + 4x$, find $p - q + r$.
A. $13x - 20y + 16z$
B. $-10y + 10$
C. $x + y + z$
D. $2x - 4y + 3x$

18. The top left corner of the figure below is a quarter of a circle. Find the perimeter of the figure below. [$\pi = 3.14$]
A. 47.42 cm
B. 42.42 cm
C. 44.57cm
D. 46.75cm



Use the diagram below to answer questions **19** *and* **20**



19. Write vector *m* as a component vector.
A. $\binom{3}{1}$
B. $\binom{-1}{3}$
C. $\binom{1}{3}$
D. $\binom{1}{-3}$

20. Find the magnitude of the vector *m*.
A. $\sqrt{5}$
B. $\sqrt{10}$
C. $\sqrt{13}$
D. $\sqrt{8}$

21. If 15% of a number is 45, find the number
A. 200
B. 250
C. 300
D. 350

22. The data **8, 9, 9, 8, 11** shows the ages (in years) of 5 learners in a class. Find the mean age of the learners.
A. 9
B. 8
C. 7.5
D. 11

23. A fair 5-sided die is tossed once. Find the probability of obtaining an odd number in decimals.
A. 0.75
B. 0.65
C. 0.8
D. 0.6

24. Evaluate 2.089 + 0.008 – 1.112
A. 0.985
B. 1.195
C. 1.005
D. 1.0

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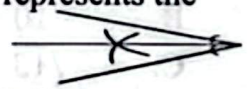
25. If $\frac{3}{4}x - \frac{1}{2} = 7$, the value of x is:
A. 2
B. 4
C. 3
D. 10

26. If 22% of a rope is 55 metres long, find the full length of the rope.
A. 12.1 m
B. 35 m
C. 121 m
D. 250 m

27. Expand $-2(3 + 2x)$
A. $5 + 4x$
B. $6 - 4x$
C. $-6 + 4x^2$
D. $-6 - 4x$

28. What is the image when a point A (4, 2) is translated by the vector $\binom{-3}{4}$?
A. $A^1(7, -2)$
B. $A^1(1, -2)$
C. $A^1(1,6)$
D. $A^1(7,6)$

29. The diagram below represents the construction for?



A. Bisection of a line segment
B. Bisecting a given angle
C. Constructing a parallel line
D. Constructing an angle of 60°

30. What is the place value of "8" in the digit 41.781?
A. Units
B. Hundredth
C. Tenth
D. Thousandth

31. Evaluate $10000^{\frac{1}{4}}$
A. 10
B. 1000
C. 100
D. 10000

32. The mean of these numbers 10, 15, x, 30 is 20. Find the missing number, x.
A. 10
B. 15
C. 25
D. 30

33. Simplify the expression $\frac{10a^4c^3}{2ac^2}$
A. $5a^2c^2$
B. $5a^4c^3$
C. $5ac$
D. $5a^3c$

34. The arithmetic mean of a set of 100 observations are 50. If each observation is increased by 2, the new mean is
A. 10
B. 50
C. 52
D. 20

35. An egg sells at 80 pesewas each. What is the cost of 1 crate of egg?
A. GH¢ 36.00
B. GH¢ 42.00
C. GH¢ 12.00
D. GH¢ 24.00

36. Find the value of y given that;
$y = (9 \div 3^3) \times 3$
A. 8
B. 2
C. 1
D. 0

37. Find the Highest Common Factor (H.C.F of 9, 18 and 30
A. 2
B. 3
C. 6
D. 9

38. Solve the equation $16^3 = 4^x$
A. 3
B. 4
C. 5
D. 6

39. Paa is 5 times as old as Eric. If the sum of their ages is 42 years, how old is Paa?
A. 30 years
B. 37 years
C. 35 years
D. 33 years

40. If $3x^2 = 75$, find x
A. $x = 3$
B. $x = 4$
C. $x = 5$
D. $x = 25$

END OF PAPER

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BEST BRAIN EXAMINATIONS KONSORTIUM
SPECIAL PRIVATE MOCK EXAMINATIONS FOR BECE CANDIDATES – SEPTEMBER 2026
MARKING SCHEME – MATHEMATICS

PAPER 2 [60 MARKS]

GENERAL NOTES ON PAPER II

1. Marks are subdivided into marks for method (M), marks for accuracy (A) and marks for accuracy not preceded by M marks (B).
2. The M marks should be given for a particular stage if the method is correct, that is, it would yield the right answer, if correctly carried out without numerical errors. M marks are not generally subdivided and unless the M mark for a preceding stage has been awarded, no A marks can be gained for that stage. (No deduction should be made from M).
3. Deduct 1 mark for omission of units or for wrong units not more than once in one whole question.
4. Give '0' (zero) for results obtained for work that is indecipherable or wholly suppressed.
5. If more questions are attempted than the rubric allows, delete the marks given for these extra questions which have the lowest marks. This rule implies that for candidates attempting more than four questions, consider only their best four and ignore the remaining questions by writing "MQA". (MQA denotes More Questions Answered than allowed)
6. Do not mark beyond the first appearance of a correct answer, i.e. ignore any further work beyond the correct answer.

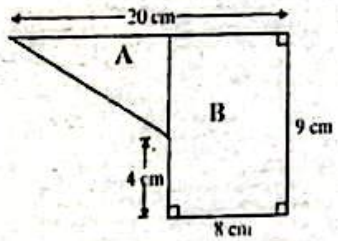
QUESTION		SOLUTION	MARKS
QUESTION 1 (a)	(i)	**Ratio of the sales made on Tuesday to Friday** Tuesday: Friday= GH¢ 700: GH¢ 1250 $= \frac{GH¢\,700}{GH¢\,1250} = \frac{70}{125}$ $= \frac{14}{25}$ ∴ 14:25	M1 *(setting up the ratio 700: 1250)* A1 *(correctly simplifying to 14:25)*
	(ii)	**Ratio of the sales made on Wednesday to Tuesday** Wednesday: Tuesday = GH¢ 1000: GH¢ 700 $= \frac{GH¢\,1000}{GH¢\,700} = 10:7$	M1 *(setting up the ratio 1000 : 700)* A1 *(correctly simplifying to 10:7)*
	(iii)	**Ratio of the sales made on Friday to total sales in the 3 days** Friday: Total sales in 3 days Total sales in 3 days = GH¢ (700+1000+1250) = GH¢ 2,950 GH¢ 1250: GH¢ 2950 GH¢ 1250 GH¢ 2950 $\frac{125}{295} = \frac{25}{59}$ 25:59	M1 *(finding total sales GH¢2,950)* M1 *(correctly forming the ratio 1250:2950 using that total)* A1 *(correctly simplifying to 25:59)*
(b)	(i)	$30 = 16 - 2x$ $\Rightarrow 2x = 16 - 30$ $\Rightarrow \frac{2x}{2} = \frac{-14}{2}$ $\Rightarrow x = -7$	M1 *(rearranging/ collecting like terms)* A1 *(correct value, x = -7)*
	(ii)	$37 - 8w = 2w - 13$ $\Rightarrow -8w - 2w = -13 - 37$	M1 *(simplification)*

		$\Rightarrow -10w = -50$ $\Rightarrow \frac{-10w}{-10} = \frac{-50}{2}$ $\Rightarrow w = 5$	A1 *(correct value, w 5)*
(c)		Fiifi completes his rounds in 36 seconds; Ama completes her rounds in 48 seconds Yaa also completes her rounds in 42 seconds. Using the L.C.M method to find the time in seconds when they will be together at the starting point. $36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$ $48 = 2 \times 2 \times 2 \times 2 \times 3 = 2^4 \times 3$ $42 = 2 \times 3 \times 7$ L. C.M $= 2^4 \times 3^2 \times 7$ L.C.M $= 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 7$ L.C.M $= 1008$ ∴ They will be together at the starting point in 1008 seconds 36 → 2, 18 → 2, 9 → 3, 3 48 → 2, 24 → 2, 12 → 2, 6 → 2, 3 42 → 2, 21 → 3, 7	M1*(correct prime factorization of all three numbers)* M1 *(correctly combining the highest powers to form the LCM expression)* A1 *(correct LCM = 1,008 seconds, stated as the final answer)*
(d)		$\frac{0.05}{100} \times 200km$ $\therefore 0.05 \times 2km = 0.1\ km$	B1 *(correct answer, 0.1km*

QUESTION 1 TOTAL = 15 MARKS

QUESTION 2			
(a)		Annual salary = GH₵ 42, 000.00, 40% = school fees, 10% = rent, 10% = tithe	
	(i)	Cost of school fees $= \frac{40}{100} \times 42{,}000 = 16{,}800.00$ ∴ **GH₵ 16,800.00**	B1 *(school fees)*
	(ii)	Cost of rent $= \frac{10}{100} \times 42{,}000 =$ **GH₵ 4,200.00**	B1 *(rent)*
	(iii)	Tithe $= \frac{10}{100} \times 42{,}000 =$ **GH₵ 4,200.00** $= 42{,}000 - (16{,}800 + 4{,}200 + 4{,}200)$ $= 42{,}000 - 25200$ N= **GH₵ 16, 800.00**	M1 *(summing the three deductions)* M1 *(subtracting from annual salary)* A2 *(correct balance)*
(b)		$4x + 2 + 8 > 30$ $4x + 10 > 30$ $4x > 30 - 10$ $4x > 20$ $x > 5$	M1 *(collecting like terms)* M1 *(isolating the x-term, 4x > 20)* A1 *(correct answer)*
(c)	(i)	4 minutes= 100 words **In 1 minute** $= \frac{100\ words}{4\ minutes} \times 1minute$ 1 minute = 25 words ∴ 25 words can be typed in 1minute	M1 A1 *(correct rate. 25 words per minute)*
	(ii)	**In 5 minutes** 5 minutes × 25 words = 125 words **OR** If 4 minutes = 100 words, then 5 minutes = x If more, less divide	B1 *(words in 5 minutes)*

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	(ii)	Average number of children in the students' families $= \frac{\Sigma fx}{\Sigma f} = \frac{56}{28} = 2$	B1 (correct mean)
(b)	(i)	$r = \frac{3}{4}$ $s = -\frac{1}{3}$ $q = r + s$ $= \frac{3}{4} + \left(-\frac{1}{3}\right)$ $= \frac{3}{4} - \frac{1}{3}$ $= \frac{9-4}{12} = \frac{5}{12}$	M1 *(correct subtraction process)* A1 *(correct final answer)*
	(ii)	$q = 3s + 2r$ $= 3\left(-\frac{1}{3}\right) + 2\left(\frac{3}{4}\right)$ $= -1 + 2\left(\frac{3}{4}\right)$ $= -1 + \frac{3}{2}$ $= \frac{-2+3}{2}$ $= \frac{1}{2}$	M1 *(correct substitution and simplification)* A1 *(final answer)*
(c)	(i)	Pattern rule = (2 × days) + 3	B1 *(correct pattern rule)*
	(ii)	Number of customers on day 11 Number of customers = (2 × days) + 3 = (2 × 11) + 3 = 22 + 3 = 25 ∴ 25 customers attended the restaurant on day 11	M½ *(substitution into the rule)* A½ *(11 days)*
	(iii)	Number of customers = (2 × days) + 3 23 = 2 days + 3 23 − 3 = 2 days $\Rightarrow \frac{20}{2} = \frac{2\ days}{2}$ $\Rightarrow$ 10 = day ∴ On day 10, the restaurant receives 23 customers.	M1 *(forming the equation 23 2(days) + 3)* A1 *(correct answer, day 10)*
(d)		 Total area = *Area of triangle(A) + Area of rectangle(B)* Base of triangle = 20 − 8 = 12 cm Height of triangle = 9 − 4 = 5cm Area of triangle = $\frac{1}{2} \times base \times height$ $= \frac{1}{2} \times 12 \times 5$ = 30 cm² Area of rectangle = *Length × Width* 9 × 8	A1 *(correct triangle area)* M1 *(rectangle area)*

		$72\ cm^2$ Total area $30 + 72$ $= 102\ cm^2$	A1 *(correct total area, 102cm²)*

QUESTION 6 TOTAL = 15 MARKS

PAPER 1[40 MARKS]

1. C	11. D	21. C	31. A
2. B	12. A	22. A	32. C
3. C	13. D	23. D	33. D
4. D	14. A	24. A	34. C
5. C	15. A	25. D	35. D
6. D	16. A	26. D	36. C
7. B	17. B	27. D	37. B
8. C	18. B	28. C	38. D
9. D	19. C	29. B	39. C
10.A	20. B	30. B	40. C

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		$x = \frac{5 \text{minutes} \times 100 \text{ words}}{4 \text{ minutes}}$ $\therefore x = 5 \times 25$ words $x = 125$ words $\therefore$ In 5 minutes, 125 words can be typed	
(d)		Length = 85m, Breadth = 25m How long the fence is = perimeter of the rectangular field Perimeter = 2l + 2B = 2 (85m) + 2 (25m) = 170m + 50m = 220m $\therefore$ The fence is 220m long	M1 A1 *(correct computation, 220m)* B1 *(correct conclusion)*
			QUESTION 2 TOTAL = 15 MARKS
QUESTION 3		$m = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ $n = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ $r = \begin{pmatrix} 18 \\ -6 \end{pmatrix}$	M1 *(adding corresponding components of the three vectors)*
(a)		$m + n + r = \begin{pmatrix} 3 \\ -1 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 18 \\ -6 \end{pmatrix}$ $= \begin{pmatrix} 3 + -1 + 18 \\ -1 + 2 + -6 \end{pmatrix}$ $= \begin{pmatrix} 3 - 1 + 18 \\ -1 + 2 - 6 \end{pmatrix}$ $m + n + r = \begin{pmatrix} 20 \\ -5 \end{pmatrix}$	A1 *(correct x-component, 20)* A1 *(correct y-component, −5)*
(b)		Perimeter of triangle = 46 mm $\Rightarrow (x + 2) + (4x - 6) + (4x - 4) = 46$ $9x - 8 = 46$ $9x = 46 + 8$ $9x = 54$ $x = \frac{54}{9} = 6$ Dimensions of the triangle $\Rightarrow (x + 2) = (6 + 2) = 8$ $4x - 6 = 4(6) - 6 = 24 - 6 = 18$ $(4x - 4) = 4(6) - 4 = 24 - 4 = 20$ $\therefore$ The dimensions of the triangle are 8 *mm*, 18 *mm* and 20 *mm*	M1 *(forming the equation)* M1 *(simplifying to 9x − 8 = 46)* A1 *(correct value, x = 6)* A1 *(correct dimensions stated)*
(c)		Length of wire = total circumference of the four semi-circles Radius of each semi-circle, $r = \frac{32}{2} = 16$ Circumference of one semi-circle = $\frac{1}{2}(2\pi r)$ $= \pi r$ $= 3.14 \times 16$ $= 50.24$ Circumference of the four semi-circles = 4×50.24 $= 200.96$ Length of the wire = 200.96 cm	M1 *(finding the radius, r = 16)* M1 *(applying the formula, πr)* A1 *(correct value 50.24cm)* A1 *(correct total for 4 semicircles)*
(d)		Mean $= \frac{\text{Total sales}}{\text{Number of days}}$ Total sales = *Mean* × *Number of days* Total sales for the first 5 days = $4.6 \times 5 = 23$ Target mean after 6 days = 5 $\Rightarrow$ Total sales after 6 days = *Mean after 6 days* × 6 $5 \times 6 = 30$ Sales needed for the 6th day for a mean of 5 *Total sales for 6 days − Total sales for 5 days* $\Rightarrow 30 - 23 = 7$ $\therefore$ The salesman must sell 7 solar power systems on the sixth day	M1*(finding total sales for first 5 days)* M1 *(sales needed after 6 days)* A1 *(subtraction)* A1 *(final answer)*
			QUESTION 3 TOTAL = 15 MARKS

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QUESTION 4 (a) (i) (i)

$y = -2x + 4$

x	-3	-2	-1	0	1	2	3	4
y	10	8	6	4	2	0	-2	-4

$y = -2x + 4$

when $x = -3$: $y = -2(-3) + 4$; $y = 6 + 4$; $y = 10$

when $x = -2$: $y = -2(-2) + 4$; $y = 4 + 4$; $y = 8$

when $x = 0$: $y = -2(0) + 4$; $y = 0 + 4$; $y = 4$

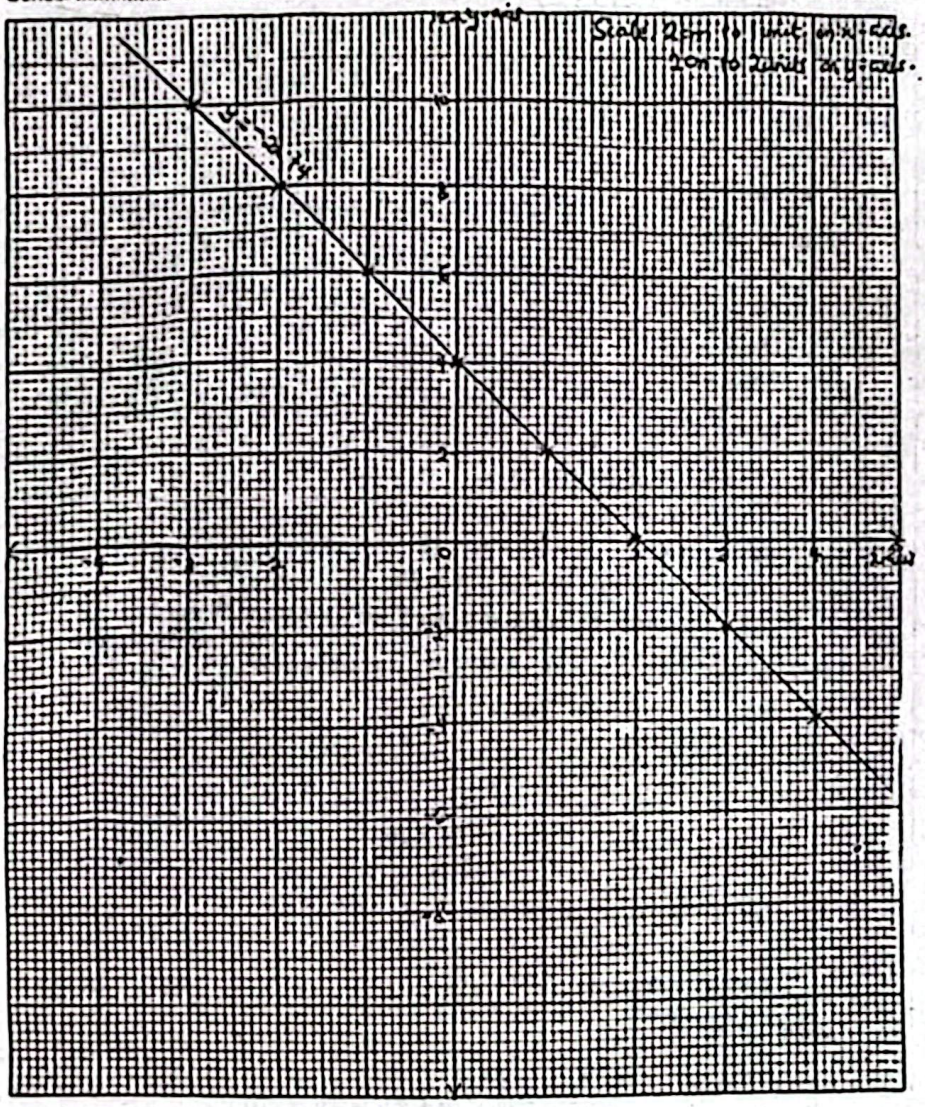
when $x = 2$: $y = -2(2) + 4$; $y = -4 + 4$; $y = 0$

when $x = 4$: $y = -2(4) + 4$; $y = -8 + 4$; $y = -4$

$B2\frac{1}{2} - \frac{1}{2}EE$

(ii, iii)

School ..... Question NO.: Aqu 4 b(ii)



Drawing Ox and Oy with correct scale and interval on each axis: $B1 - \frac{1}{2}EE$

Plotting and indicating the coordinates of all points: $B3 - \frac{1}{2}EE$

Drawing straight lines to join points for the line: $B1\frac{1}{2} - \frac{1}{2}EE$

Penalties

Wrong/non-labeling of vertices: $-\frac{1}{2}EE$

Non-joining or non-use of straight edge: $-\frac{1}{2}EE$

Non-calibration of axes: $-\frac{1}{2}EE$

Non-labelling of axes: $-\frac{1}{2}EE$

(b)		48 hours in seconds 3600 seconds = 1hour 48 hours = 48 × 3600s 48 hours = 172800 seconds **In Words; One hundred and seventy – two thousand, eight hundred seconds**	B$\frac{1}{2}$ B$\frac{1}{2}$ A$\frac{1}{2}$ A$\frac{1}{2}$
(c)		$\|5(7-3)-4(5-8)\|$ $\|5(4)-4(-3)\|$ $\|20+12\|$ 32	M1 A1
(d)		In the nearest tenth = 48.3 To the nearest hundredth = 48.29 To the nearest thousandth = 48.294	A1 A1 A1

QUESTION 4 TOTAL = 15 MARKS

QUESTION 5 (a)	(i)	$\overrightarrow{OA} = \begin{pmatrix}3\\1\end{pmatrix}$ $\overrightarrow{OB} = \begin{pmatrix}-1\\-2\end{pmatrix}$	B1 *(both position vectors stated)*
	(ii)	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $= \begin{pmatrix}-1\\-2\end{pmatrix} - \begin{pmatrix}3\\1\end{pmatrix}$ $= \begin{pmatrix}-4\\-3\end{pmatrix}$	M1 *(vector subtraction)* A1 *(correct result, AB = (–4, –3))*
	(iii)	$\|\overrightarrow{AB}\| = \sqrt{(-4)^2 + (-3)^2}$ $= \sqrt{16+9}$ $= \sqrt{25}$ $= 5$	M1 *(magnitude formula)* A1 *(correct magnitude)*
(b)	(i)	(i) $5\frac{3}{4} - 2\frac{7}{8} + 1\frac{1}{2}$ $= \frac{23}{4} - \frac{23}{8} + \frac{3}{2}$ $= \frac{46-23+12}{8}$ $= \frac{35}{8}$ $= 4\frac{3}{8}$	M1 *(converting to a common denominator across three mixed numbers)* A1 *(final answer*
	(ii)	$\frac{8}{9} \times \frac{4}{6} = \frac{4}{9} \times \frac{4}{3} = \frac{16}{27}$	B1 (correctly simplified)
(c)		Fraction given to his daughter = $\frac{1}{3}$ Fraction given to his first son = $\frac{1}{5}$ Fraction given to his wife = $\frac{2}{5}$ Total Fraction spent = $\frac{1}{3} + \frac{1}{5} + \frac{2}{5}$ $= \frac{5+3+6}{15}$ $= \frac{14}{15}$ Fraction Left = 1 – Fraction spent $= \frac{1}{1} - \frac{14}{15}$ $= \frac{15-14}{15}$ Fraction Left = $\frac{1}{15}$	M1 *(adding the three fractions over a common denominator to get 14/15 spent)* A1 *(correct fraction left, 1/15, following directly from the total spent)*
(d)		**Angle of sector for Aged Parent** 42° + 60° + 65° + 78° + 60° + Aged Parent = 360° 305° + Aged Parent = 360°	M1 *(finding the Aged Parent's*

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Aged Parent = 360 – 305°
Aged Parent = 55°

angle (360° 305° 55°)

(i) **Amount he earns in a month**

If 55° = GH₵ 66.00

then 360° = x, where x is the amount, he earns

$\frac{55° \times x}{55°} = \frac{360° \times ₵\ 66.00}{55°}$

$x = \frac{360 \times ₵6.00}{5}$

$x = 72 \times$ GH₵ 6.00

x = GH ₵ 432.00

∴ He earns GH₵ 432.00 in a month

M1 *(setting up the proportion for monthly earnings (55° GH₵66 → 360° x))*

A1 *(monthly earnings)*

(ii) **Amount spent on others**

$\frac{42°}{360°} \times$ GH₵ 432.00

$\frac{42}{30} \times ₵\ 36.00$

$\frac{42 \times ₵12}{10}$ = GH₵ 50.40

A1 *(correct amount spent on Others)*

(iii) **Percentage of his salary on Health and Food**

$\frac{60° + 78°}{360°} \times 100\%$

$\frac{138°}{360} \times 100\%$

$\frac{138 \times 10\%}{36}$

38.33%

A1 *(correct percentage on Health and Food)*

QUESTION 5 TOTAL = 15 MARKS

QUESTION 6

(a) (i)

Number of children in student families (x)	Tally	Frequency (f)	F(x)
1	𝍸 𝍸	10	10
2	𝍸 𝍸 //	12	24
3	////	4	12
4	/	1	4
6	/	1	6
	Totals	28	56

$B3 - \frac{1}{2}EE$

(correct table and F(x) values and column)

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